SECTION 5: SCIENCE PROFICIENCY
Time -- 25 minutes
50 Questions (181 - 230)

NAME: _____
SCHOOL: _____
SIMULATED EXAM - SET B

For each item, decide which the best answer choice is, and darken the corresponding circle on the answer sheet.

181. Assume that an Rh- mother has given birth to an Rh+ baby and was not given RhoGAM. When might hemolytic disease of the newborn occur?
   A. Hemolytic disease of the newborn will never occur
   B. At the birth of the first Rh- baby
   C. All pregnancies will show hemolytic disease
   D. During development of the next Rh+ baby
   E. Hemolytic disease will never occur to an Rh+ baby from Rh- mother is normal.

182. In a certain characteristic of garden peas, tall is dominant over short. If two heterozygous plants are crossed, what percentage of the next generation will be homozygous dominant?
   A. 0%
   B. 25%
   C. 50%
   D. 75%
   E. 100%

183. Which among the following is TRUE in the ABO blood group?
   A. Type A blood has Type A antigen and Type B antibody.
   B. Type A blood has Type B antigen and Type B antibody.
   C. Type A blood has Type A antigen and Type B, O antibodies.
   D. Type A blood has Type B, O antigens and Type A antibody.
   E. Type A blood has both Type A antigens and antibody.

184. In which of the following vessels does most of the exchange of nutrients and waste products between the blood and body tissues happens?
   A. arteries
   B. capillaries
   C. veins
   D. lymphatic vessels
   E. renal tubules

185. Which substance is released in order to expose binding sites for muscular contraction?
   A. actin
   B. myosin
   C. calcium
   D. troponin
   E. ATP

186. What process involves the enzymatic breakdown of food?
   A. Absorption
   B. Digestion
   C. Mastication
   D. Peristalsis
   E. Segmentation

187. Which organelle is found in animal cells but *not* in plant cells?
   A. ribosomes
   B. mitochondria
   C. nuclei
   D. plastids
   E. centrioles

188. After telophase II of meiosis, the chromosomal makeup of each daughter cell is _____.
   A. diploid, and the chromosomes are each composed of a single chromatid.
   B. diploid, and the chromosomes are each composed of two chromatids.
   C. haploid, and the chromosomes are each composed of a single chromatid.
   D. haploid, and the chromosomes are each composed of two chromatids.
   E. haploid, and the chromosomes are each composed of half of the chromatid

189. What is the important animal hormone responsible for the control of metabolic functions in the blood particularly for iodine synthesis?
   A. thyroxine
   B. cortisol
   C. prostaglandin
   D. insulin
   E. parathormone

190. What phylum is commonly known as round worms?
   A. Mollusca
   B. Echinodermata
   C. Platyhelminthes
   D. Nematoda
   E. Porifera

191. A sexually reproducing animal has two unlinked genes, one for head shape (H) and one for tail length (T). Its genotype is HhTt. Which of the following parent genotypes is a possible gamete to form this organism?
   A. Hhtt x hhtt
   B. HhTT x HHTT
   C. HhTt x HhTt
   D. HHTT x HHTt
   E. hhTt x hhtt

192. DNA is to _____ as proteins is to _____ .
   A. nucleus . . . amino acid
   B. nucleotides . . . amino acid
   C. nitrogenous bases . . . peptide
   D. nucleus . . . peptide
   E. RNA . . . genotype

193. In a particular plant, leaf color is controlled by gene locus D. Plants with at least one allele D have dark green leaves, and plants with the homozygous recessive dd genotype have light green leaves. A true-breeding dark-leaved plant is crossed with a light-leaved one, and the F_1 offspring is allowed to self-pollinate. The predicted outcome of F_2 is diagrammed in the Punnett square shown in the figure below where 1, 2, 3, and 4 represent the genotypes corresponding to each box within the square.

	D	d
D	1	2
d	3	4

Which of the boxes marked 1-4 correspond to plants with homozygous genes?
   A. 1 and 2
   B. 2 and 3
   C. 3 and 4
   D. 1 and 4
   E. 2, 3 and 4

194. Tissues are composed of cells, and tissues functioning together make up organs. Which of the following is composed of different organs?
   A. organelles
   B. organ membranes
   C. organ system
   D. organisms
   E. cytoplasm

195. Which part of the flower will develop into seeds?
A. stigma
B. stamen
C. ovary
D. ovule
E. anther

196. Matter consists of _____.
A. atom
B. molecule
C. cell
D. organelle
E. tissue

197. Matter is defined as anything that has _____.
A. mass and shape
B. mass and density
C. mass only
D. mass and volume
E. volume and energy

198. If 25 mL of water at 40°C is added to 100 mL of water at 50°C, what would be the final temperature of the mixture?
A. 56°C
B. 48°C
C. 42°C
D. 44°C
E. 88°C

199. Thirty liters of gas X at 100 K under 2 atm was transferred to a certain container where the temperature is doubled and the pressure is increased by 4 atm. What is the mass of the gas if its density at the new container is 0.01 g/L?
A. 20 g
B. 10 g
C. 0.2 g
D. 0.05 g
E. 2000 g

200. Two hundred milliliters of water is mixed with 720 grams of HCl. What is the molarity of the solution if the density of water is 1.00 grams per milliliter? (Cl= 35 amu, H=1 amu)
A. 100 M
B. 0.278 M
C. 0.1 M
D. 129.6 M
E. 20 M

201. Consider the following reaction:

$$AlCl_3 + Ca(OH)_2 \rightarrow Al(OH)_3 + CaCl_2$$

How many moles of aluminum hydroxide is produced when 10 moles of aluminum chloride is reacted to 12 moles of calcium hydroxide?
A. 10 moles
B. 8 moles
C. 20 moles
D. 12 moles
E. 4 moles

202. What are the representative elements in the periodic table?
A. Elements that belong to IIIB to IIB.
B. Elements that have last sublevel configuration of s or p.
C. Elements that have f sublevel configuration.
D. Elements that belong to IA–VIIIA and IIIB to IIB.
E. Elements that have sublevel configuration of s, p, d and f blocks

203. Which of the following is a double displacement reaction?
A. $2HCl + Mg \rightarrow MgCl_2 + H_2$
B. $2AlPO_4 + 3Ca \rightarrow Ca_3(PO_4)_2 + 2Al$
C. $NaOH + HCl \rightarrow NaCl + H_2O$
D. $2KClO_3 \rightarrow 2KCl + 3O_2$
E. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

204. How much water must be added to 300 mL of 0.75 M HCl to dilute the solution to 0.25 M HCl solution?
A. 100 mL
B. 300 mL
C. 600 mL
D. 900 mL
E. 1200 mL

205. Copper has two isotopes, ^{63}Cu and ^{65}Cu. If the atomic mass of copper is 63.5 g/mol, what is the % of each isotope?
A. 25.0% ^{63}Cu and 75.0% ^{65}Cu
B. 33.0% ^{63}Cu and 67.0% ^{65}Cu
C. 50.0% ^{63}Cu and 50.0% ^{65}Cu
D. 67.0% ^{63}Cu and 33.0% ^{65}Cu
E. 75.0% ^{63}Cu and 25.0% ^{65}Cu

206. What is the value of X in the reaction below if the equation would be balanced?

$$C_4H_6 + O_2 \rightarrow XCO_2 + H_2O$$

A. 2
B. 4
C. 6
D. 7
E. 8

207. What is the limiting reagent for the following chemical reaction if 4 grams of hydrogen and 32 grams of oxygen react to form water?

$$2H_2 + O_2 \xrightarrow{electricity} 2H_2O$$

A. hydrogen
B. oxygen
C. water
D. this reaction has no limiting reagent
E. electricity

208. Given the following:

I. $^{40}_{20}X^{2+}$
II. $^{40}_{21}X^{2+}$
III. $^{40}_{20}X$
IV. $^{39}_{20}X$
V. $^{39}_{19}X^{2+}$

Which of the following pairs of element is an isotope?
A. I and II
B. I and III
C. II and III
D. III and IV
E. IV and V

209. A nitride ion N^{3-} (Z=7) has _____.
A. 7 protons and 3 electrons
B. 7 protons and 4 electrons
C. 7 protons and 10 electrons
D. 10 protons and 7 electrons
E. 4 protons and 7 electrons

210. Which of the following equations is ***not*** balanced correctly?
A. $P_4 + 3O_2 \rightarrow P_4O_6$
B. $3CuCl_2 + V_2O_3 \rightarrow 3CuO + 2VCl_3$
C. $Cu(NO_3)_2 \rightarrow CuO + 2NO_2 + O_2$
D. $2KClO_3 \rightarrow 2KCl + 3O_2$
E. $2Al_2O_3 + 3C + 6Cl_2 \rightarrow 4AlCl_3 + 3CO_2$

211. Which of the following is a definition of a base?
A. It is a proton donor.
B. It turns blue litmus paper to red.
C. It is usually formed when a metal oxide reacts with water.
D. Its pOH is above 7.
E. It is an electron pair acceptor.

212. The periodic trend that refers to the ability of an atom to attract electrons during chemical bonding is _____.
   A. electronegativity
   B. electron affinity
   C. atomic radius
   D. ionization energy
   E. radioactivity

213. A graduated cylinder has a mass of 80.0 g when empty. When 20.0 mL of water is added, the graduated cylinder has a mass of 100. g. If a stone is added to the graduated cylinder, the water level rises to 45.0 mL and the total mass is now 156 g. What is the density of the stone?
   A. 2.24 g/mL
   B. 3.47 g/mL
   C. 1.56 g/mL
   D. 2.60 g/mL
   E. 7.80 g/mL

214. Which of the following is an expected property of sugar ($C_6H_{12}O_6$)?
   A. It does not turn to black when heated.
   B. It has melting point greater than 100°C.
   C. It is insoluble in water at 35°C.
   D. Chemical change occurs when it is heated at 80°C.
   E. It can be filtered when dissolved in alcohol.

215. An atom has 7 electrons in its outer shell. How many unpaired electrons does it have?
   A. 0
   B. 1
   C. 2
   D. 4
   E. 6

216. A vector quantity has magnitude and direction. Which of the following is a vector quantity?
   A. speed
   B. gravitational pull
   C. work
   D. distance
   E. temperature

217. Given

   Vector A = 15 units, west
   Vector B = 8 units, north

   Find the magnitude of the resultant vector.
   A. 7 units
   B. 17 units
   C. 23 units
   D. 24 units
   E. 25 units

218. A ball was thrown vertically upward with a speed of 5 m/s. When the ball went back to where it was thrown, what was its displacement?
   A. 10 meters
   B. 5 meters, upward
   C. zero
   D. 5 meters, downward
   E. half a meter

219. A projectile is launched at a certain angle with the ground at a speed of 20 meters per second. Neglecting air resistance, what is the speed of the projectile when it hits the ground?
   A. 9.8 meters per second
   B. 10 meters per second
   C. zero
   D. 20 meters per second
   E. 25 meters per second

220. If acceleration is the rate of change in velocity, what is the rate of change in position?
A. Velocity
B. Distance
C. Force
D. Displacement
E. Speed

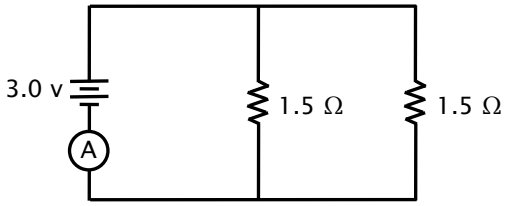
221. A golf ball, initially resting on the tee, was hit by a club. Where in its path, did the golf ball have maximum kinetic energy?
A. Just right after it was hit by the club
B. A few seconds after it was hit by the club
C. On the way up
D. On the way down
E. At the highest point

222. Two objects, X and Y, are being pulled at the same speed. Object X weighs 500 kg and object Y weighs 2000 kg. What is the ratio of the kinetic energy of object Y to the kinetic energy of object X?
A. 1:1
B. 2:1
C. 4:1
D. 9:1
E. 16:1

223. A 50-kilogram girl lifts a 1-kilogram object vertically upward to a height of 0.5 meters. How much work is done by the girl in lifting the object?
A. 5 joules
B. 50 joules
C. 250 joules
D. 10 joules
E. 500 joules

224. Which of the following describes electrical current?
A. the electric potential difference between two points in a circuit
B. the flow of electric charge through a material
C. the flow of protons through a material
D. the movement of voltage
E. the movement of electric field through a conductor

225. Refer to the circuit below. What would be the reading in the ammeter?

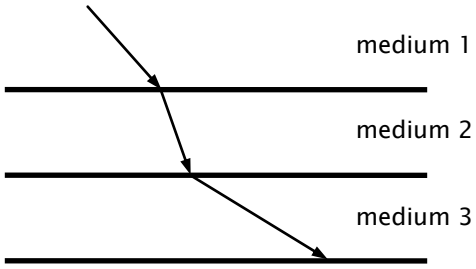


A. 3.0 amperes
B. 1.5 amperes
C. 4.5 amperes
D. 4.0 amperes
E. 1.0 ampere

226. What power is supplied to the starter motor of a car that draws 240 amperes of current from a 12-volt battery?
A. 2.0×10^1 watts
B. 5.4×10^3 watts
C. 3.5×10^3 watts
D. 6.0×10^1 watts
E. 4.5×10^3 watts

227. A wave that moves from left to right, creating an up and down displacement of the medium, is a
A. longitudinal wave.
B. transverse wave.
C. mechanical wave.
D. electromagnetic wave.
E. sound wave.

228. Refer to the figure below. A ray of light passes through three different media. Which of the following is correct about the speed of the light in these media?



A. $v_1 > v_2 > v_3$
B. $v_2 > v_1 > v_3$
C. $v_2 > v_3 > v_1$
D. $v_3 > v_2 > v_1$
E. $v_3 > v_1 > v_2$

229. A piece of lead tied by string to a spring scale weighs 63 N. When the piece of lead is completely submerged in a certain liquid, the reading on the spring scale is 57 N. What would the reading in the spring scale be if the liquid were replaced by a twice-as-dense liquid?
A. 51 N
B. 114 N
C. 126 N
D. 54 N
E. 60 N

230. In uniform circular motion, what is the relationship between acceleration and velocity?
A. Acceleration is directly proportional to velocity.
B. Acceleration is inversely proportional to velocity.
C. Acceleration is directly proportional to the square of the velocity
D. Acceleration is inversely proportional to the square of the velocity
E. Acceleration is directly proportional to the square root of the velocity